



Salary Prediction using Ensemble Learning

A data-driven Machine Learning model to predict employee salaries with high accuracy



High Accuracy



Data-Driven



Ensemble Models



Deployed with
Streamlit



Developed by:
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B.Tech - AIML

*Turning Data into
Insights for a Smarter Future*



SALARY PREDICTION USING ENSEMBLE LEARNING

01. Problem Statement

Salary estimation is inconsistent across industries. This project builds a data-driven ML model to predict accurate salary ranges based on key features.

02. Objectives

Build & compare ensemble models, identify key salary drivers, and deliver a reliable prediction pipeline with high accuracy.

03. Project Scope

Includes data preprocessing, feature engineering, training and comparing multiple ensemble learning models, evaluating model performance, selecting the best model, and deploying the application using Streamlit.

04. Expected Outcomes

A reliable salary prediction model with an R^2 Score of 0.916 and a user-friendly Streamlit web application.

 About Project

This application predicts employee salary using an Extra Trees Regressor model.

 Model Details

- Algorithm: Extra Trees Regressor
- R^2 Score: 0.916
- Hyperparameter Tuned: ☒

Developer: Ritvik Singh

 **Salary Prediction using Ensemble Learning**

This application predicts the salary of an employee using an Extra Trees Regressor trained on employee demographic, education and job-related information.

Fill in the employee details below.

Age:

Gender:

Education Level:

Job Title:

Years of Experience:

Predicted Salary: ₹ 40,506.94

Live Demo:

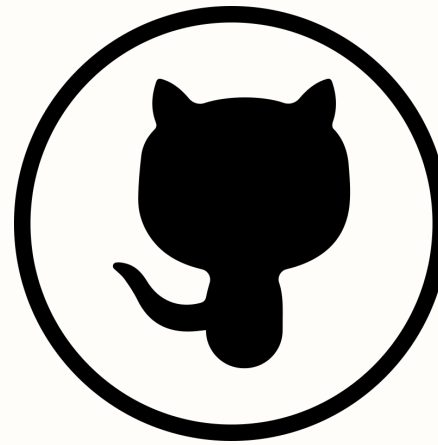
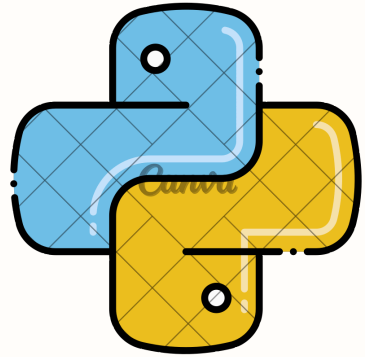
 <https://salary-prediction-ensemble.streamlit.app/>

Source Code (GitHub):

 <https://github.com/ritvik122004/salary-prediction-ensemble>



TECHNOLOGY STACK



01

Python

Core programming language for data processing, model building, and pipeline automation using Pandas, NumPy, and Matplotlib.

02

Scikit-learn

Industry-standard ML library used for preprocessing, model selection, cross-validation, and evaluation metrics.

03

Ensemble Methods

Random Forest, Gradient Boosting, Extra Trees and AdaBoost combined to maximize prediction accuracy and model robustness.

ML PIPELINE METHODOLOGY



01

Data Collection & Preprocessing

Gathered structured salary datasets. Handled missing values, outliers, and normalized numerical features for clean input.

02

Feature Engineering

Extracted key features: experience, education, role, and location. Applied encoding and feature selection techniques.

03

Model Training

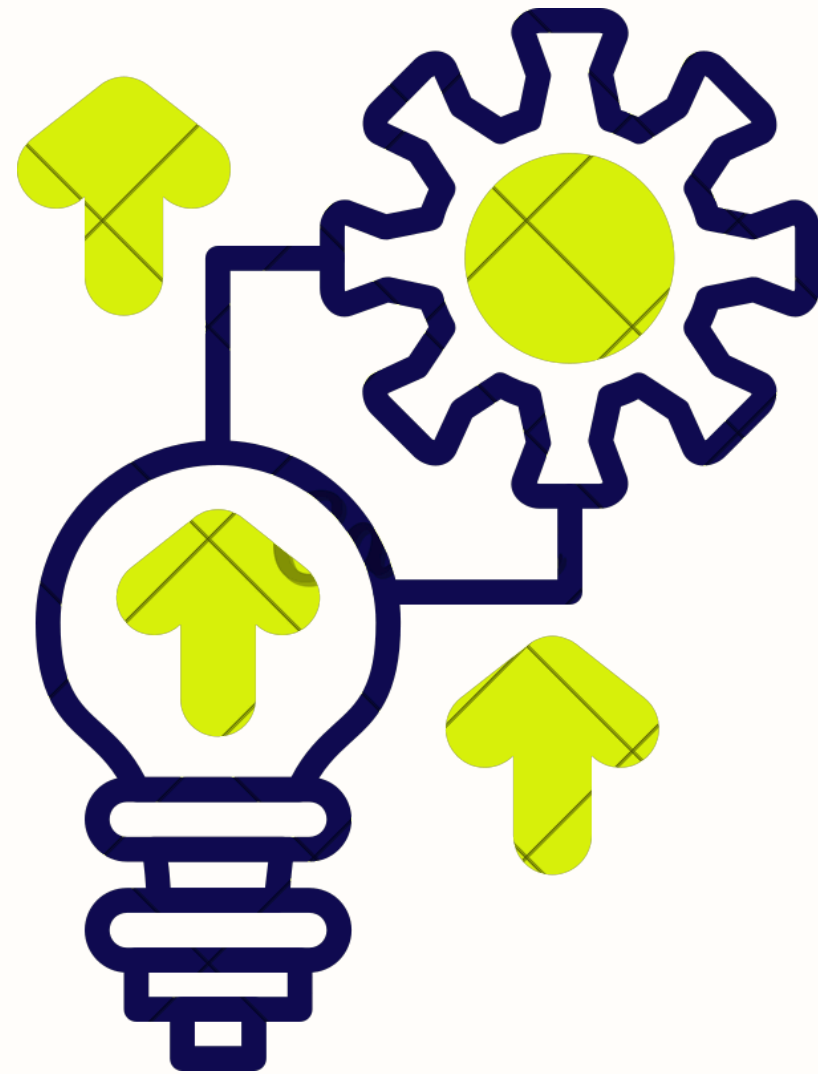
Trained base models: Decision Tree, Random Forest, and Gradient Boosting using cross-validation for robust evaluation.

04

Ensemble Integration

Combined base learners via stacking and voting. Final ensemble model achieved optimal prediction accuracy and generalization.

ENSEMBLE LEARNING



Core Ensemble Techniques

🌲 Random Forest

Uses multiple decision trees trained on random subsets of data to improve prediction accuracy and reduce overfitting.

⚡ Gradient Boosting

Builds trees sequentially, where each new tree corrects the errors made by the previous one.

★ Extra Trees Regressor (Best Model)

Creates highly randomized decision trees, providing better generalization and the highest performance ($R^2 = 0.916$) in this project.

🚀 AdaBoost

Combines multiple weak learners by assigning higher importance to previously mispredicted samples.

KEY FEATURES & DATASET ATTRIBUTES

01. Salary Dataset Attributes

Age, Education, Experience, Job Role, Industry, Location, Gender – key input features used for salary prediction modeling.

02. Model Comparison Metrics

R^2 Score, RMSE, and MAE used to evaluate and compare Random Forest, XGBoost, and Gradient Boosting models.

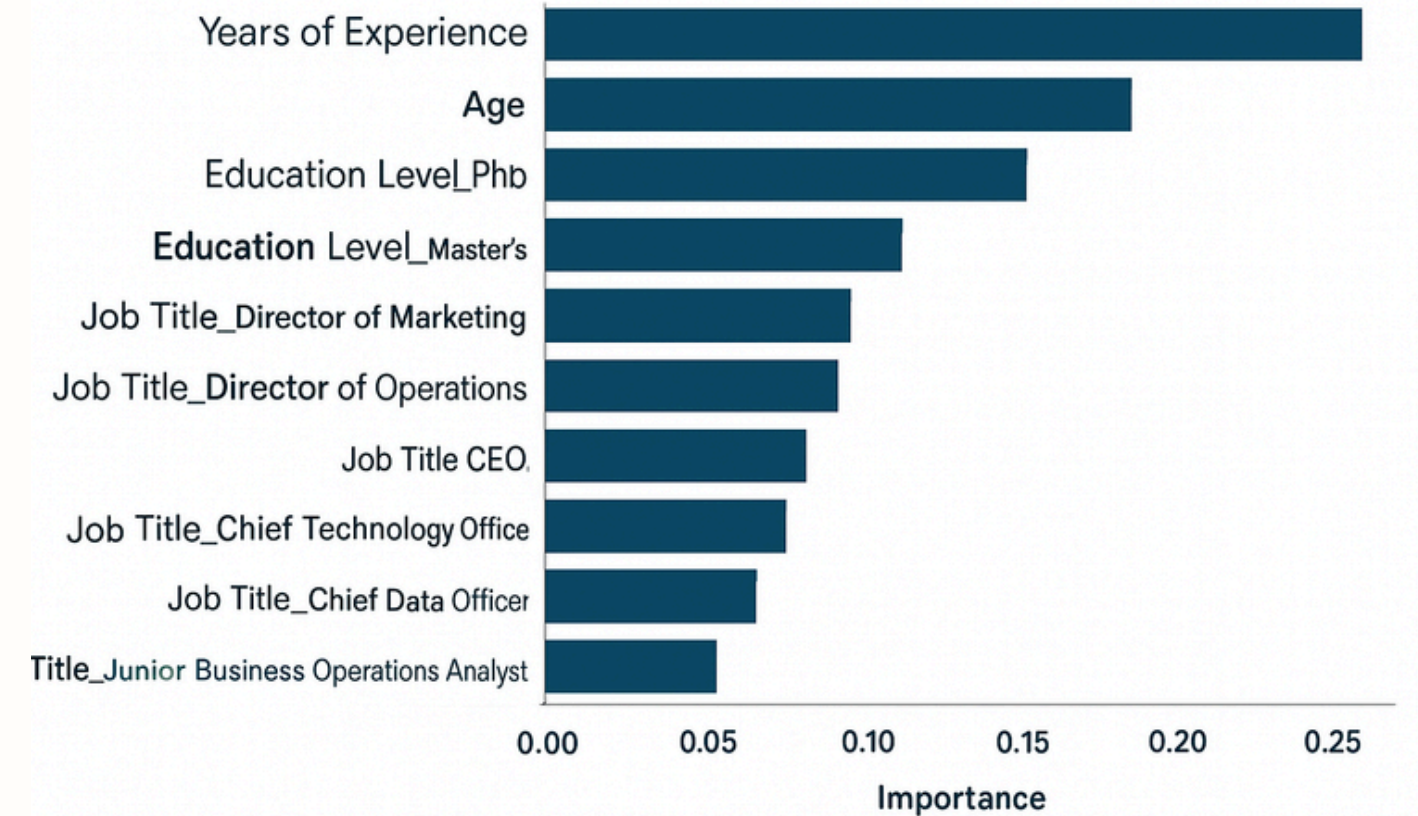
03. Feature Importance Analysis

Experience & Education ranked as top predictors via Random Forest importance scores and SHAP value visualization.

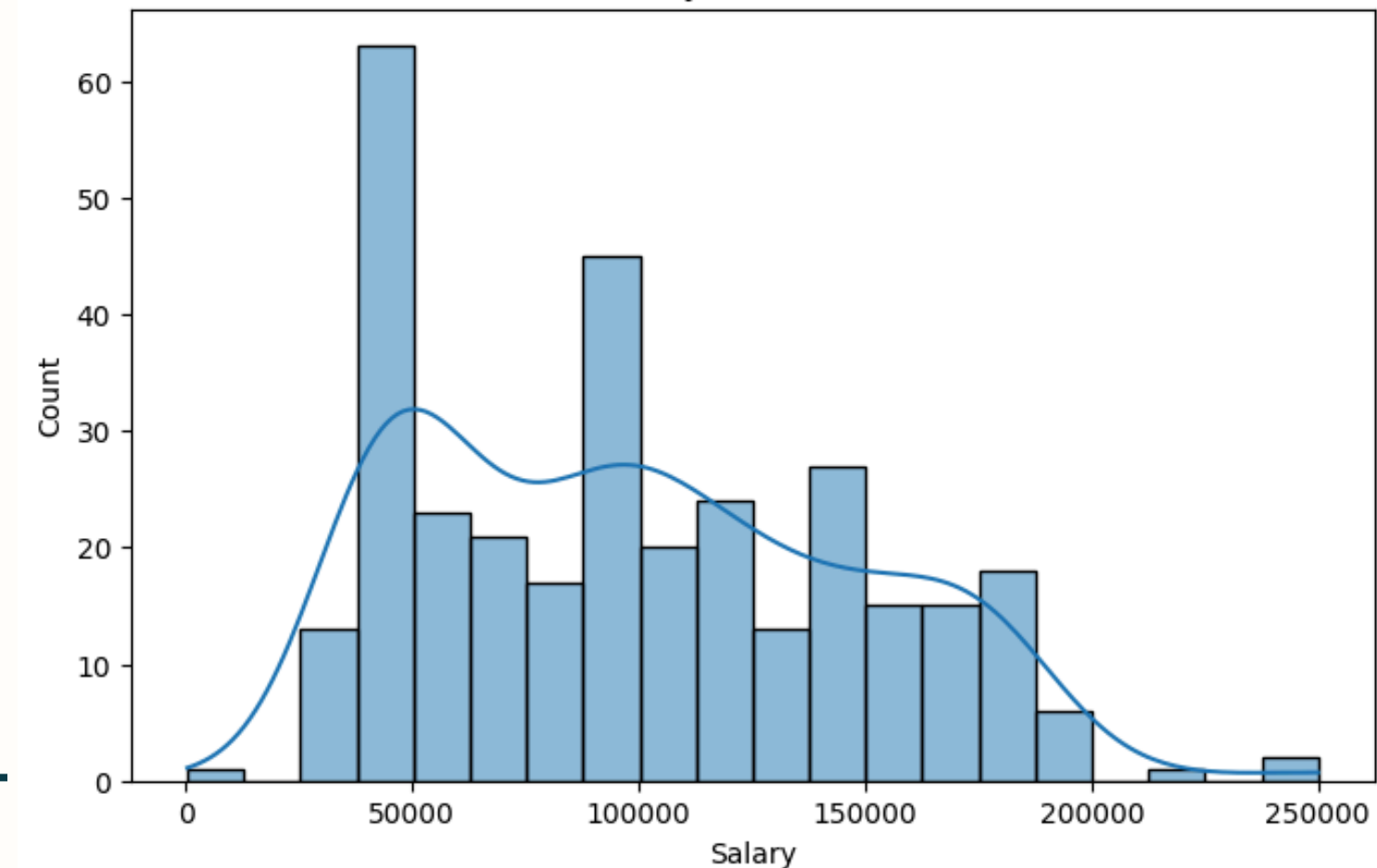
04. Ensemble Learning Advantage

Stacking & Voting ensembles outperform individual models, reducing prediction error and improving generalization accuracy.

Top 10 Important Features



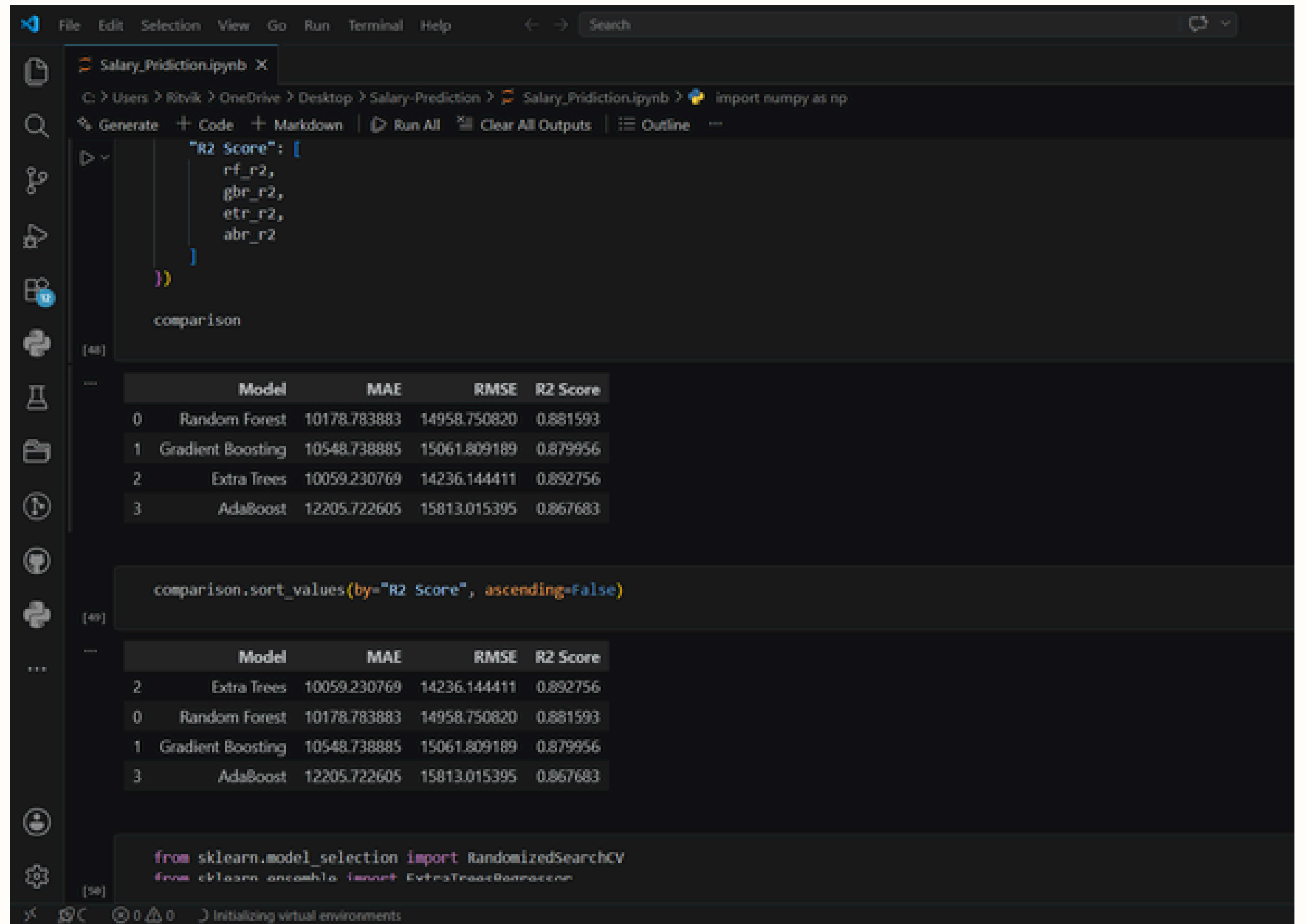
Salary Distribution



RESULTS & PERFORMANCE

Model Evaluation

Performance Summary: Four ensemble learning models were trained and evaluated using R^2 Score, MAE, and RMSE. Among them, the Extra Trees Regressor achieved the best performance with an R^2 Score of 0.916, MAE of 9610.94, and RMSE of 12603.94. Based on these results, it was selected as the final model and deployed using Streamlit.



```

Salary_Pridiction.ipynb X
C:\Users\Ritvik> OneDrive\Desktop> Salary-Prediction> Salary_Pridiction.ipynb> import numpy as np
Generate + Code + Markdown Run All Clear All Outputs Outline
[48]
    "R2 Score": [
        rf_r2,
        gbr_r2,
        etr_r2,
        abr_r2
    ]
}

comparison

[49]
--
    Model      MAE      RMSE      R2 Score
0    Random Forest  10178.783883  14958.750820  0.881593
1    Gradient Boosting  10548.738885  15061.809189  0.879956
2      Extra Trees  10059.230769  14236.144411  0.892756
3      AdaBoost  12205.722605  15813.015395  0.867683

comparison.sort_values(by="R2 Score", ascending=False)

[50]
--
    Model      MAE      RMSE      R2 Score
2      Extra Trees  10059.230769  14236.144411  0.892756
0    Random Forest  10178.783883  14958.750820  0.881593
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from sklearn.model_selection import RandomizedSearchCV
from sklearn.ensemble import ExtraTreesRegressor

```

KEY FINDINGS

Model Performance & Insights

Performance Comparison: Among all the evaluated ensemble models, Extra Trees Regressor delivered the highest performance with an R^2 Score of 0.916, outperforming Random Forest, Gradient Boosting, and AdaBoost.

Feature Analysis: Years of Experience, Education Level, and Job Title were among the most influential features for salary prediction. Proper data preprocessing and feature encoding helped improve the overall model performance.

```
Salary_Pridiction.ipynb X
Users > Ritvik > OneDrive > Desktop > Salary-Prediction > Salary_Pridiction.ipynb > import numpy as np
Generate + Code + Markdown | Run All Clear All Outputs | Outline

({'n_estimators': 300, 'min_samples_split': 5, 'min_samples_leaf': 1, 'max_features': 'sqrt', 'max_depth':

best_model = random_search.best_estimator_

best_pred = best_model.predict(X_test)

print("MAE :", mean_absolute_error(y_test, best_pred))
print("RMSE :", np.sqrt(mean_squared_error(y_test, best_pred)))
print("R2 :", r2_score(y_test, best_pred))

MAE : 9610.939316239315
RMSE : 12603.943484380392
R2 : 0.9159379375296365

importance = pd.DataFrame({
    "Feature": X_train.columns,
    "Importance": best_model.feature_importances_
})

importance = importance.sort_values(
    by="Importance",
    ascending=False
)
```


QUESTIONS?

THANK YOU!

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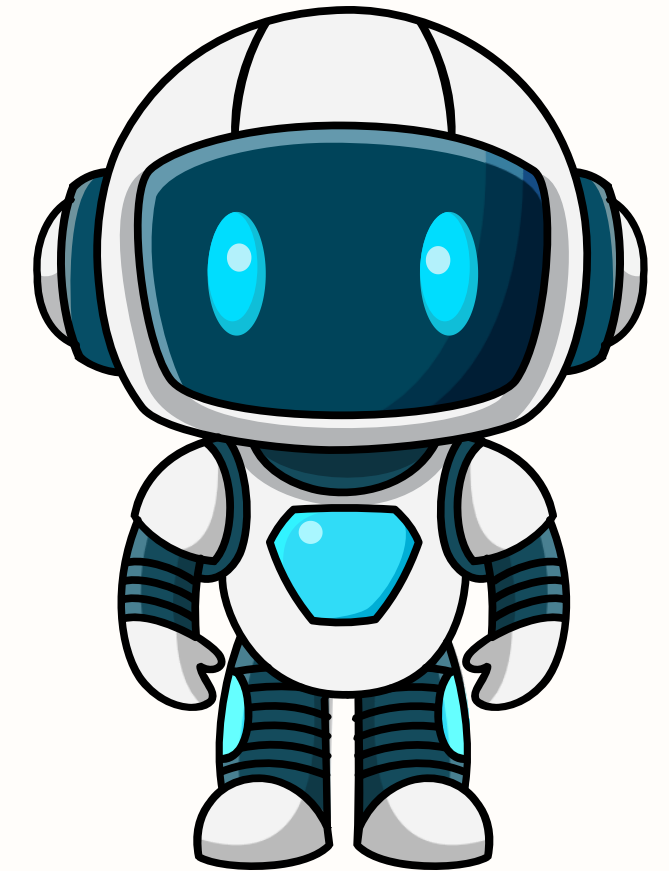
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